Supplementary Information

Vacuum-Transition-Induced Spacetime Throats: Gravitational Collapse, Matter Transfer, and Conditional Cosmological Rebirth

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S1. Equations and scheme

The metric is $ds^2=-f\,du\,dv+r^2d\Omega^2$ with $f=e^{\sigma}$, $G_{\mu\nu}=T_{\mu\nu}$ ($8\pi G=1$) and $T_{\mu\nu}=\partial_\mu\varphi\partial_\nu\varphi-g_{\mu\nu}[(\nabla\varphi)^2/2+U]$. A symbolic computation of the Einstein tensor (derive_equations.py) gives

$$r_{uu}-\sigma_u r_u=-\frac{r}{2}\phi_u^2, \qquad r_{uv}=-\frac{r_u r_v}{r}-\frac{f}{4r}+\frac{Ufr}{4} \tag{S1}$$

$$\phi_{uv}=-\frac{\phi_u r_v+\phi_v r_u}{r}-\frac{f}{4}U'(\phi), \qquad \sigma_{uv}=\frac{f+4r_u r_v}{2r^2}-\phi_u\phi_v \tag{S2}$$

The first of Eqs. (S1) is the constraint along u (that along v is Eq. (4) of the paper) and the others are the evolution equations. The scheme advances rows of constant u on a uniform lattice $u_j=jh$, $v_i=ih$. A cell with corners $W=(j,i-1)$, $A=(j+1,i-1)$, $B=(j,i)$ gives the new point $C=(j+1,i)$ by $X_C=X_A+X_B-X_W+h^2X_{uv}$, with X_{uv} evaluated at the cell centre from the mean values and the central differences, iterated three times (predictor–corrector). Near the centre the evolution of σ divides by a small radius and its rounding noise is amplified, so within $K=6$ points of the centre σ is integrated from the constraint $\sigma_v=(r_{vv}+r\varphi_v^2/2)/r_v$ (trapezoid), and beyond it from the evolution equation. Each row is solved with the centre value of σ as the unknown: elementary flatness requires $\sigma_c=\ln(4r_v^2)$ at the centre, which defines a residual that is solved by the secant method (ten passes at most); an iteration on the plain residual has a gain of modulus greater than one and diverges. The centre value of φ follows from $\varphi_t=2\varphi_v$ at the centre by an implicit trapezoid with the one-sided derivative (the explicit iteration has gain -1.5).

The data on the initial cone are $r=v/2$, $\varphi(0,v)=A\psi(v)$ and $\sigma(0,v)=\mathrm{int}_0^v r\varphi_v^2dv$, the integral of the constraint along v with $f=1$ at the centre. The potential $U=(\kappa/4)(\psi^2-1)^2+s\psi+d$ is tuned by a root finder so that the false minimum has density 3 and the true minimum $-3q$.

S2. Numerical diagnostics

Misner–Sharp mass of exact solutions: Table 2 of the paper. The diagnostics of each run are the signs of r_u (a forward difference in u) and r_v (a central difference in v) at every point more than four cells from the centre and eight from the outer boundary; the quadrant map of Fig. 4 is produced by the same classification.

S3. Thin-wall check

Table S0 lists the check of the critical mass. For each of nine combinations of q and $\hat{\sigma}$ the exact value $m_c=\max_R m_+(R)$ is computed on a grid of two million points, and the existence of a forbidden window is tested at $m_c(1\mp10^{-3})$.

q	$\hat{\sigma}$	m_c	Window at $m_c(1-10^{-3})$	Window at $m_c(1+10^{-3})$
0.5	0.01	1.49999	yes	no
0.5	0.30	1.48757	yes	no
0.5	0.90	1.39332	yes	no

q	$\hat{\sigma}$	m_c	Window at $m_c(1-10^{-3})$	Window at $m_c(1+10^{-3})$
1.0	0.01	1.99998	yes	no
1.0	0.30	1.98505	yes	no
1.0	0.90	1.86927	yes	no
2.0	0.01	2.99998	yes	no
2.0	0.30	2.98253	yes	no
2.0	0.90	2.84482	yes	no

Table S0. Check of the critical mass (units of R_{dS}).

S4. Scan of the collapse runs

Supplementary Table S1 lists all runs: field amplitude A, wall width w, core radius $R_w=V_w/2$, exterior mass m_{ext} (standard deviation across the exterior in parentheses), $R_M=(3m_{ext}/\Delta)^{1/3}$, the radius of the first trapped point, the horizon radius r_h of the measured mass, and the number of anti-trapped points in the core on the last row.

A	w	R_w	m_{ext}	R_M	r (first trapped)	r_h	Anti-trapped core points
0.05	0.05	0.40	0.166 (0.000)	0.437	0.163	0.162	0
0.05	0.05	0.60	0.514 (0.000)	0.636	0.429	0.433	0
0.05	0.05	0.80	1.158 (0.000)	0.834	0.743	0.745	0
0.05	0.05	1.00	2.188 (0.001)	1.030	1.044	1.045	0
0.05	0.05	1.20	3.687 (0.001)	1.226	1.330	1.331	31
0.05	0.05	1.50	7.001 (0.001)	1.518	1.738	1.739	127
0.05	0.05	2.00	16.033 (0.003)	2.001	2.387	2.390	287
0.05	0.05	3.00	51.525 (0.007)	2.953	3.627	3.632	607
0.05	0.10	0.40	0.198 (0.000)	0.462	—	0.191	0
0.05	0.10	0.60	0.589 (0.001)	0.665	0.474	0.479	0
0.05	0.10	0.80	1.301 (0.001)	0.867	0.791	0.796	0
0.05	0.10	1.00	2.426 (0.002)	1.066	1.095	1.099	0
0.05	0.10	1.20	4.054 (0.003)	1.266	1.382	1.387	0
0.05	0.10	1.50	7.638 (0.006)	1.563	1.795	1.801	59
0.05	0.10	2.00	17.393 (0.011)	2.056	2.455	2.462	159
0.05	0.10	3.00	55.941 (0.030)	3.036	3.725	3.737	359
0.05	0.20	0.40	0.324 (0.001)	0.545	—	0.298	0
0.05	0.20	0.60	0.932 (0.003)	0.775	0.626	0.653	0
0.05	0.20	0.80	2.000 (0.006)	1.000	0.968	1.000	0
0.05	0.20	1.00	3.656 (0.009)	1.223	1.285	1.326	0
0.05	0.20	1.20	6.042 (0.013)	1.446	1.590	1.639	0
0.05	0.20	1.50	11.381 (0.021)	1.785	2.039	2.101	19
0.05	0.20	2.00	48.383 (4.594)	2.892	—	3.552	119
0.05	0.20	3.00	110.863 (8.554)	3.813	—	4.735	319
0.10	0.05	0.40	0.191 (0.000)	0.457	0.183	0.185	0
0.10	0.05	0.60	0.554 (0.000)	0.652	0.456	0.458	0
0.10	0.05	0.80	1.194 (0.000)	0.842	0.755	0.758	0
0.10	0.05	1.00	2.179 (0.000)	1.029	1.041	1.043	0
0.10	0.05	1.20	3.569 (0.001)	1.213	1.310	1.312	31
0.10	0.05	1.50	6.535 (0.001)	1.484	1.690	1.692	127
0.10	0.05	2.00	14.238 (0.002)	1.924	2.285	2.286	287
0.10	0.05	3.00	42.413 (0.007)	2.768	3.389	3.392	607
0.10	0.10	0.40	0.203 (0.000)	0.467	—	0.196	0
0.10	0.10	0.60	0.592 (0.001)	0.666	0.480	0.481	0
0.10	0.10	0.80	1.285 (0.001)	0.863	0.789	0.791	0
0.10	0.10	1.00	2.362 (0.002)	1.057	1.084	1.085	0
0.10	0.10	1.20	3.900 (0.003)	1.249	1.362	1.364	0
0.10	0.10	1.50	7.226 (0.006)	1.534	1.760	1.761	59
0.10	0.10	2.00	16.052 (0.011)	2.002	2.389	2.391	159
0.10	0.10	3.00	49.421 (0.030)	2.913	3.575	3.579	359
0.10	0.20	0.40	0.255 (0.000)	0.503	—	0.241	0
0.10	0.20	0.60	0.712 (0.002)	0.709	0.545	0.548	0

A	w	R_w	m_{ext}	R_M	r (first trapped)	r_h	Anti-trapped core points
0.10	0.20	0.80	1.511 (0.004)	0.911	0.864	0.865	0
0.10	0.20	1.00	2.739 (0.007)	1.110	1.163	1.164	0
0.10	0.20	1.20	4.480 (0.010)	1.308	1.447	1.447	0
0.10	0.20	1.50	8.238 (0.017)	1.603	1.854	1.855	19
0.10	0.20	2.00	18.222 (0.032)	2.089	2.504	2.505	119
0.10	0.20	3.00	56.409 (0.084)	3.044	3.744	3.748	319
0.20	0.05	0.40	0.271 (0.000)	0.514	0.248	0.254	0
0.20	0.05	0.60	0.671 (0.000)	0.695	0.524	0.526	0
0.20	0.05	0.80	1.289 (0.000)	0.864	0.790	0.792	0
0.20	0.05	1.00	2.151 (0.000)	1.025	1.033	1.037	0
0.20	0.05	1.20	3.286 (0.001)	1.180	1.262	1.264	31
0.20	0.05	1.50	5.567 (0.001)	1.407	1.584	1.585	127
0.20	0.05	2.00	11.217 (0.003)	1.777	2.089	2.090	287
0.20	0.05	3.00	31.893 (0.007)	2.517	3.066	3.066	607
0.20	0.10	0.40	0.249 (0.000)	0.499	—	0.236	0
0.20	0.10	0.60	0.661 (0.001)	0.692	0.511	0.520	0
0.20	0.10	0.80	1.340 (0.001)	0.875	0.805	0.810	0
0.20	0.10	1.00	2.331 (0.002)	1.052	1.074	1.078	0
0.20	0.10	1.20	3.674 (0.003)	1.225	1.326	1.329	0
0.20	0.10	1.50	6.426 (0.005)	1.476	1.679	1.681	59
0.20	0.10	2.00	13.259 (0.011)	1.879	2.226	2.226	159
0.20	0.10	3.00	37.225 (0.029)	2.650	3.241	3.239	359
0.20	0.20	0.40	0.282 (0.000)	0.521	0.248	0.264	0
0.20	0.20	0.60	0.750 (0.001)	0.721	0.553	0.567	0
0.20	0.20	0.80	1.535 (0.003)	0.916	0.864	0.872	0
0.20	0.20	1.00	2.700 (0.006)	1.105	1.150	1.156	0
0.20	0.20	1.20	4.306 (0.009)	1.291	1.419	1.423	0
0.20	0.20	1.50	7.655 (0.015)	1.564	1.800	1.802	19
0.20	0.20	2.00	16.144 (0.030)	2.006	2.395	2.396	119
0.20	0.20	3.00	46.330 (0.078)	2.851	3.501	3.499	319

Table S1. Collapse runs (q=1, κ=24).